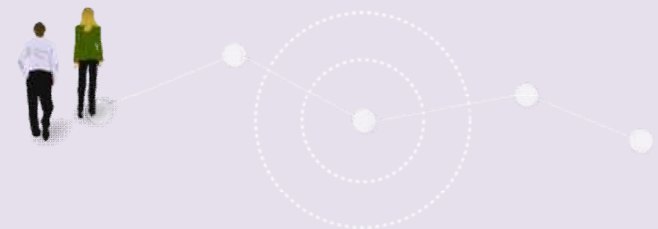


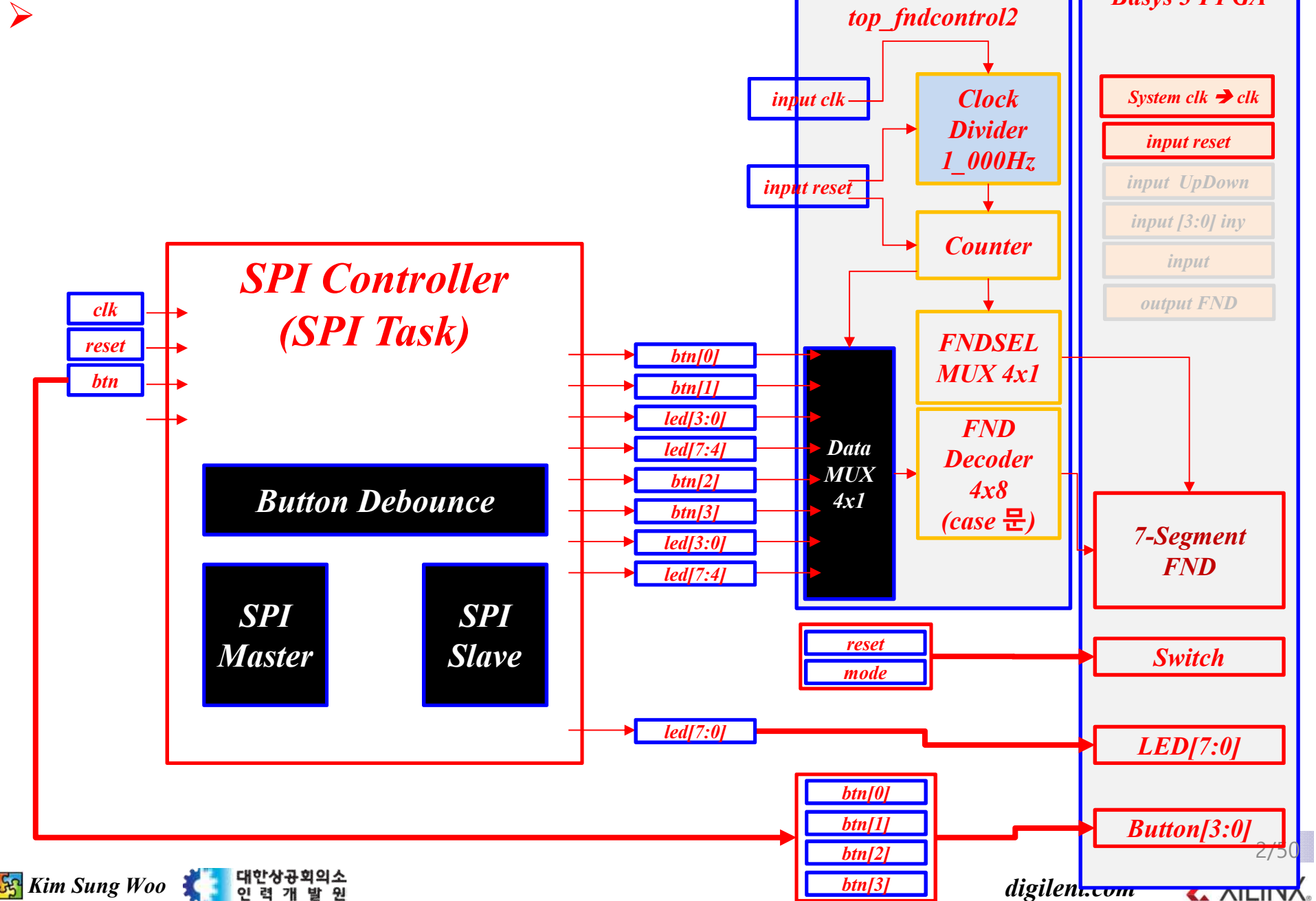
Example 00. SPI_Controller_Exam_1

➤ *spi_controller_exam_1.xpr*



Example 00. SPI_Controller_Exam_1

➤ spi_controller_exam_1.xpr



Example 00. SPI_Controller_Exam_1

➤ spi_controller_exam_1.xpr

- btn[0] → BTNR (T18)
- btn[1] → BTND (W19)
- btn[2] → BTNU (T17)
- btn[3] → BTNU (U17)
- reset → SW15
- clk → System Clock [100MHz]
- FND4 → rdata 10 (led[7:4])
- FND3 → rdata 0(led[7:4])
- FND2 → Mode 0 : btn[1]
- FND1 → Mode 0 : btn[0]
- FND2 → Mode 1 : btn[3]
- FND1 → Mode 1 : btn[2]
- LD 7 ~ LD 0 → led_out [7:0]



실습

```
1. ## Clock signal
2. set_property -dict { PACKAGE_PIN W5  IOSTANDARD LVCMOS33 } [get_ports clk]
3. create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]
4. ## Switches
5. set_property -dict { PACKAGE_PIN T1  IOSTANDARD LVCMOS33 } [get_ports {mode}]
6. set_property -dict { PACKAGE_PIN R2  IOSTANDARD LVCMOS33 } [get_ports {reset}]
7. ## LEDs
8. set_property -dict { PACKAGE_PIN U16  IOSTANDARD LVCMOS33 } [get_ports {led_out[0]}]
9. set_property -dict { PACKAGE_PIN E19  IOSTANDARD LVCMOS33 } [get_ports {led_out[1]}]
10. set_property -dict { PACKAGE_PIN U19  IOSTANDARD LVCMOS33 } [get_ports {led_out[2]}]
11. set_property -dict { PACKAGE_PIN V19  IOSTANDARD LVCMOS33 } [get_ports {led_out[3]}]
12. set_property -dict { PACKAGE_PIN W18  IOSTANDARD LVCMOS33 } [get_ports {led_out[4]}]
13. set_property -dict { PACKAGE_PIN U15  IOSTANDARD LVCMOS33 } [get_ports {led_out[5]}]
14. set_property -dict { PACKAGE_PIN U14  IOSTANDARD LVCMOS33 } [get_ports {led_out[6]}]
15. set_property -dict { PACKAGE_PIN V14  IOSTANDARD LVCMOS33 } [get_ports {led_out[7]}]
16. ##7 Segment Display
17. set_property -dict { PACKAGE_PIN W7  IOSTANDARD LVCMOS33 } [get_ports {fnd[0]}]
18. set_property -dict { PACKAGE_PIN W6  IOSTANDARD LVCMOS33 } [get_ports {fnd[1]}]
19. set_property -dict { PACKAGE_PIN U8  IOSTANDARD LVCMOS33 } [get_ports {fnd[2]}]
20. set_property -dict { PACKAGE_PIN V8  IOSTANDARD LVCMOS33 } [get_ports {fnd[3]}]
21. set_property -dict { PACKAGE_PIN U5  IOSTANDARD LVCMOS33 } [get_ports {fnd[4]}]
22. set_property -dict { PACKAGE_PIN V5  IOSTANDARD LVCMOS33 } [get_ports {fnd[5]}]
23. set_property -dict { PACKAGE_PIN U7  IOSTANDARD LVCMOS33 } [get_ports {fnd[6]}]
24. #set_property -dict { PACKAGE_PIN V7  IOSTANDARD LVCMOS33 } [get_ports fnd[7]]
25. set_property -dict { PACKAGE_PIN U2  IOSTANDARD LVCMOS33 } [get_ports {fndsel[0]}]
26. set_property -dict { PACKAGE_PIN U4  IOSTANDARD LVCMOS33 } [get_ports {fndsel[1]}]
27. set_property -dict { PACKAGE_PIN V4  IOSTANDARD LVCMOS33 } [get_ports {fndsel[2]}]
28. set_property -dict { PACKAGE_PIN W4  IOSTANDARD LVCMOS33 } [get_ports {fndsel[3]}]
29. ##Buttons
30. #set_property -dict { PACKAGE_PIN U18  IOSTANDARD LVCMOS33 } [get_ports btnC]
31. set_property -dict { PACKAGE_PIN T18  IOSTANDARD LVCMOS33 } [get_ports btn[0]]
32. set_property -dict { PACKAGE_PIN W19  IOSTANDARD LVCMOS33 } [get_ports btn[1]]
33. set_property -dict { PACKAGE_PIN T17  IOSTANDARD LVCMOS33 } [get_ports btn[2]]
34. set_property -dict { PACKAGE_PIN U17  IOSTANDARD LVCMOS33 } [get_ports btn[3]]
```

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Example 00. SPI_Controller_Exam_1

➤ spi_controller_exam_1.xpr

- *btn[0]* ➔ BTNR (T18)
- *btn[1]* ➔ BTND (W19)
- *btn[2]* ➔ BTNU (T17)
- *btn[3]* ➔ BTNU (U17)
- *reset* ➔ SW15
- *clk* ➔ System Clock [100MHz]
- *FND4* ➔ *rdata 10 (led[7:4])*
- *FND3* ➔ *rdata 0 (led[7:4])*
- *FND2* ➔ *Mode 0 : btn[1]*
- *FND1* ➔ *Mode 0 : btn[0]*
- *FND2* ➔ *Mode 1 : btn[3]*
- *FND1* ➔ *Mode 1 : btn[2]*
- *LD 7 ~ LD 0* ➔ *led_out [7:0]*

